

cs2521

Week 3 | Sorting

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Agenda

- housekeeping
- Sorting algorithm properties
 - stability
 - adaptability
 - in-place

housekeeping

- engagement marks are now given on tutorials
- lectures are a prerequisite for tutorials
- tutorials will consist of a (brief) recap and then groupwork

Sorting properties

- Preserves relative order between items of equal comparisons

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- Lets try seeing this in action
- <https://cgi.cse.unsw.edu.au/~cs2521/26T3/tutorials/week03/stable-sorter.html>

Adaptability

- Adaptive algorithms change behaviour/complexity based off the input data
- Consider the following two arrays:

Array 1: [5, 1, 2, 3, 4]

Array 2: [3, 1, 5, 4, 2]

Lets try sorting this on the whiteboard using bubble sort

Adaptability

- Adaptive algorithms change behaviour/complexity based off the input data
- Consider the following two arrays:

Array 1: [5, 1, 2, 3, 4]

Array 2: [3, 1, 5, 4, 2]

Lets try sorting this on the whiteboard using bubble sort

- As we can see, sorting the first array is faster than sorting array 2
- Thats adaptability!

In place

Sorting properties

- In place algorithms conduct the sorting within the original structure
- This saves us space complexity!
- Lets take a look at selection sort & mergesort

In place

- In place algorithms conduct the sorting within the original structure
- This saves us space complexity!
- Lets take a look at selection sort & mergesort
- mergesort needs a copy of the array
- selection sort sorts on the same algorithm

Tutorial time!
